MOAIN AL-ABBASI

Software Engineer · Full-Stack Systems · AI Agents (LangGraph, MCP)

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SUMMARY

Software engineer building business systems and AI agents end to end in TypeScript, Python and Go: database, API, web and mobile clients, hardware integration and deployment. Work includes a POS that runs on an existing Oracle database and prints to Bluetooth thermal printers, an attendance system that pulls directly from ZKTeco devices and exports Saudi WPS payroll files, a multi-tenant notification platform with an Android SMS gateway, and Basira, an Arabic quotation verifier with 0/500 false alarms and 487 automated tests. Fully remote, same time zone as Saudi Arabia.

EXPERIENCE

Founder & Software Engineer · Alabbasi Tech (alabbasi.uk) · Sana'a, Yemen · Remote — 2026 – Present

- **POS replacing YemenSoft Onyx Pro** — web POS that works directly on the existing Oracle database: sales, returns, shifts and reports. Arabic receipts print on a Bixolon SPP-R310 over Bluetooth SPP through Web Serial; documented why the Xprinter USB path fails on Windows and the supported alternatives. — NestJS 10, React 19, Oracle PL/SQL, BullMQ, OpenAPI
- **Motech Attendance** — replaces the ZKTeco desktop software: branch collectors pull punches from iFace950 devices over TCP into a central multi-tenant service with idempotent ingest; shifts, leave balances, CSV/Excel reports and WPS payroll export for Saudi Arabia (MHRSD/SAMA) and the UAE (SIF). Tested on a live device (11 employees, 3,118 punches). — Go, PostgreSQL, Python (pyzk), systemd
- **Nexus Notify** — multi-tenant, multi-channel notification platform behind one API, with an Angular dashboard and a native Android SMS gateway (dual SIM, foreground service). — NestJS, PostgreSQL 17, Prisma, Redis/BullMQ, Angular 22, Kotlin, Jetpack Compose
- **Secure remote access** — Zero-Trust platform with signed agents, per-user policies and session audit; motech-cli is a cross-platform single-command agent that joins the NetBird mesh, with cosign-signed releases built in GitHub Actions. — Go, SSH, NetBird
- **Google Ads tooling for Keif Al-Diafa (Jeddah)** — persistent OAuth connector to Google Ads and Search Console with free-form GAQL queries and an Arabic dashboard; phase 1 of a multi-client ads platform with live Google Ads API data and a 12-table schema. — Hono, Cloudflare D1, Next.js 15, Prisma, PostgreSQL 16
- **keif-publisher** — generates posts from a media archive, converts video to 9:16, routes every post through a human approval gate, then publishes on schedule; dry-run and kill switch on by default. — Python, FastAPI, APScheduler
- Websites for Saudi hospitality brands: keifaldiafa.com, asoulaldiafa.com, qahwajeen.com. In development: an offline-first event-management PWA (Angular 22, IndexedDB) and a modern layer around a legacy PHP accounting system.

AI AGENTS & APPLIED AI

Basira · basirapp.site · github.com/MoTechSys/Project-Basira
Verifies Quran and Hadith quotations byte-exactly against 71,987 records from licensed corpora; an optional LLM only proposes spans and reads images (OCR), and never decides a verdict. 150/150 evaluation cases, 0/500 false alarms, 78.54% on IslamicEval 2025 subtask 1B (public dev set). 487 automated tests (345 backend, 110 bot, 32 frontend); ruff, mypy strict, Playwright with axe in CI. REST API, MCP server and Telegram bot. Entry to the AI in Service of Islamic Content Challenge 2026. — Python, FastAPI, React 19

Mansur AI (private repository)
Arabic multi-agent system: planner, ReAct executor, verifier and learner orchestrated with LangGraph; 13 agents, 38 tools, model router with fallback and cost tracking, hierarchical memory (working, episodic, procedural, semantic) and skills saved from successful runs. Sandboxed shell and SSRF-protected HTTP tools; 82 passing tests. — Python, LangGraph, FastAPI, WebSocket, SQLite

Ai_Alabbasi
Coding agent with a model backend swapped from one config file (Claude, GPT-5, DeepSeek or a local model): planner with a task list, state machine with interrupts, CodeAct sandboxed Python (local, Daytona or E2B), hybrid retrieval with FTS5 and vectors fused by RRF.

Moain Promoter AI
Seven specialised agents under an orchestrator that write bilingual Arabic/English posts for six platforms, with pluggable model providers and a Next.js dashboard. — Python, Celery, Redis, Playwright

MCP and agent tooling
Diafa Studio (Arabic PDF generation exposed to AI agents through MCP), a Telegram MCP server, and Bina, reusable agents for website audit, rebuild and Arabic SEO content.

SELECTED PROJECTS

Idhaat · github.com/MoTechSys/Idhaat-Platform
Live-tutoring platform: conflict-free allocation of four Zoom rooms with fallback to self-hosted Jitsi secured by a JWT for each lesson; recordings upload to Cloudflare R2 in chunks during class, play in a protected player with a moving watermark of the student's name, and are deleted after 48 hours. Homework, messaging and course finances; runs for about 25 SAR a month. — Hono, Cloudflare Pages, D1, R2, Vitest

Grocery Ledger · github.com/MoTechSys/mohanad-web-app-2
Offline Android accounting and POS app for small shops: barcode by camera (ML Kit) or USB/Bluetooth scanner, append-only ledgers, PDF/Excel export. 157 passing tests, v2.2.1. — Flutter

Sijil · github.com/MoTechSys/Sijilati
Offline debt ledger for shop owners and workers, including those who cannot read: amounts entered as banknotes, spoken confirmation, 8-second undo. 93 passing tests. — Flutter

Ghaida · github.com/MoTechSys/ghaida
Household guidebook for domestic workers in their native language with AI voice for every sentence; offline PWA and admin panel; API tests 27/27, end-to-end journey 17/17.

S-ACM · github.com/MoTechSys/S-ACM-Project
Academic content-management system with AI summarisation and question generation. — React 19, TypeScript, Tailwind 4

GRADUATION PROJECT (TEAM)

Security Operations Center (SOC) · github.com/MoTechSys/my-bro
SOC built on open-source tools (Wazuh 4.14.7, Suricata, YARA, VirusTotal, auditd, MikroTik syslog) with an AI analyst whose actions require human approval. 14 use cases verified on a running Wazuh 4.14.7 lab (2026-09-28), including automatic blocking of SSH brute-force attacks. Hands-on digital forensics lab: Autopsy, FTK Imager, registry, event logs and browser artefacts.

TECHNICAL SKILLS

Languages	TypeScript, Python, Go, Dart, Kotlin, SQL, PL/SQL
Backend	NestJS, FastAPI, Hono, Prisma, REST/OpenAPI, WebSockets, Redis, BullMQ, Celery
Frontend & mobile	React 19, Next.js, Angular 22, Tailwind CSS, PWA, IndexedDB, Arabic RTL; Flutter, Jetpack Compose, ML Kit
Data	PostgreSQL, Oracle, SQLite, Cloudflare D1
AI & agents	LangGraph, multi-agent orchestration, MCP servers, RAG, hybrid search (FTS5, vectors, RRF), model routing; OpenAI, Anthropic, Gemini
Integration	ZKTeco (pyzk), ESC/POS over Bluetooth and Web Serial, Android SMS gateway, Google Ads API, Search Console
Security	Wazuh, Suricata, YARA, auditd; Autopsy, FTK Imager; NetBird, cosign
DevOps & quality	Docker, GitHub Actions, Cloudflare (Pages, D1, R2), Linux, systemd; pytest, Vitest, Playwright, ruff, mypy strict

LANGUAGES

Arabic: native · English: technical reading and writing